ASSIGNMENT 14.3

2303A51335
Karthik Reddy
Batch 20

Task 1: AI-Assisted Personal Portfolio Website
Scenario
A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.
Tasks
1. Use GitHub Copilot to generate a semantic HTML structure with sections:
o Home
o About
o Projects
o Contact
2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.
3. Customize AI-generated CSS to add:
o Hover effects on project cards
o Smooth scrolling navigation
Expected Outcome
• Clean and well-structured portfolio website
• Responsive layout for desktop and mobile screens
• Interactive project cards

PROMPT:
<!--Generate a semantic HTML structure with the following sections: Home- About Projects Contact Then, suggest responsive CSS using Flexbox or Grid for layout.customize the CSS to include:Hover effects on project cards (e.g., scale up slightly, add shadow) Smooth scrolling navigation between sections--!>
<!--Ensure the HTML uses semantic tags (header, nav, section, footer) and the CSS is clean, modern, and responsive.--!>

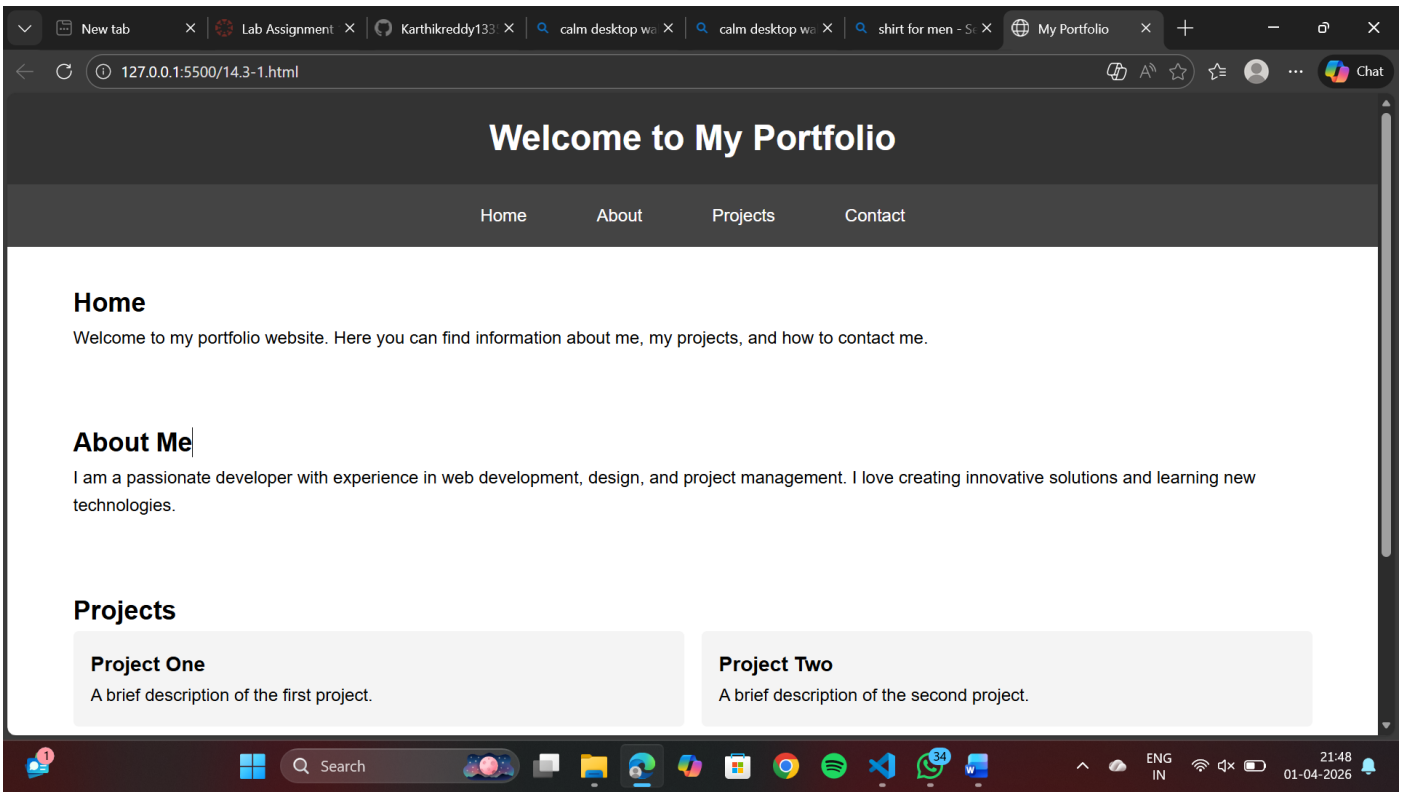
CODE:
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio</title>
  <style>
    /* Basic Reset */
    * {
      margin: 0;
```

```
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  line-height: 1.6;
}

header {
  background: #333;
  color: #fff;
  padding: 1rem 0;
  text-align: center;
}

nav {
  display: flex;
  justify-content: center;
  background: #444;
}

nav a {
  color: #fff;
  padding: 1rem 2rem;
  text-decoration: none;
  transition: background 0.3s ease;
}

nav a:hover {
  background: #555;
}

section {
  padding: 2rem;
  max-width: 1200px;
  margin: auto;
}

.projects {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 1rem;
}
```

```
      .project-card {
        background: #f4f4f4;
        padding: 1rem;
        border-radius: 5px;
        transition: transform 0.3s ease, box-shadow 0.3s ease;
      }

      .project-card:hover {
        transform: scale(1.05);
        box-shadow: 0 4px 8px rgba(0,0,0,0.2);
      }

      footer {
        background: #333;
        color: #fff;
        text-align: center;
        padding: 1rem 0;
      }
    </style>
  </head>
  <body>
    <header>
      <h1>Welcome to My Portfolio</h1>
    </header>
    <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
    <section id="home">
      <h2>Home</h2>
      <p>Welcome to my portfolio website. Here you can find information about me, my projects, and
how to contact me.</p>
    </section>
    <section id="about">
      <h2>About Me</h2>
      <p>I am a passionate developer with experience in web development, design, and project
management. I love creating innovative solutions and learning new technologies.</p>
    </section>
    <section id="projects">
      <h2>Projects</h2>
      <div class="projects">
        <div class="project-card">
          <h3>Project One</h3>
```

```
        <p>A brief description of the first project.</p>
      </div>
      <div class="project-card">
        <h3>Project Two</h3>
        <p>A brief description of the second project.</p>
      </div>
      <!-- Add more project cards as needed -->
    </div>
  </section>
  <section id="contact">
    <h2>Contact Me</h2>
    <p>You can reach me at
      <a href="mailto:example@email.com">example@email.com</a>
    </p>
  </section>
  <footer>
    <p>&copy; 2023 My Portfolio. All rights reserved.</p>
  </footer>
</body>
</html>
```

OUTPUT:



Task 2: AI-Generated Restaurant Landing Page

Scenario
A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.
Tasks
1. Use Copilot to create a navigation bar with links:
o Home
o Menu
o Gallery
o Contact
2. Generate a menu section styled with CSS cards.
3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.
Expected Outcome
• Fully functional restaurant landing page
• Responsive navigation bar
• Working image slider with smooth transitions

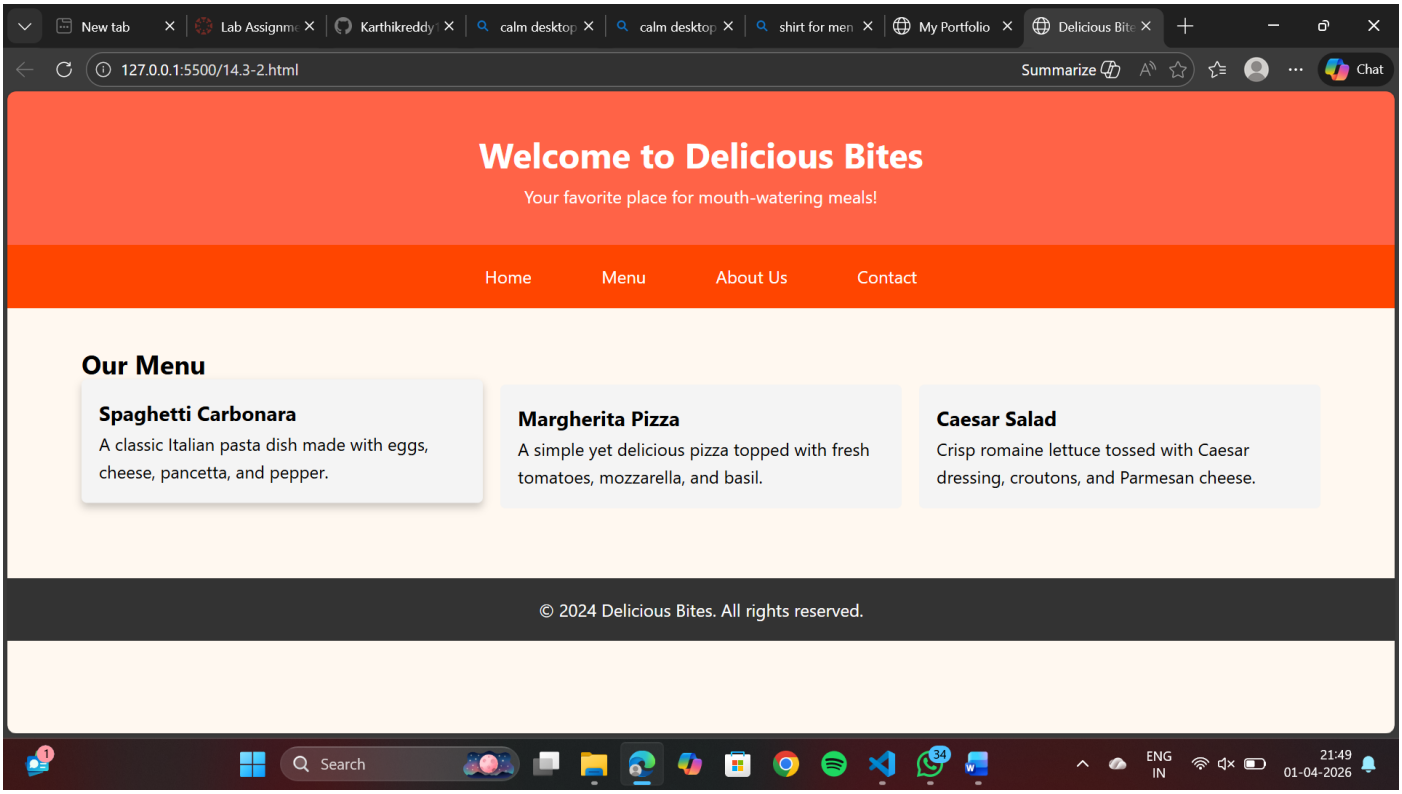
PROMPT:
<!-- create a basic HTML5 structure for a restaurant landing page -->

CODE:
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Delicious Bites - Restaurant Landing Page</title>
  <style>
    /* Basic Reset */
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      line-height: 1.6;
      background-color: #fff8f0;
    }

    header {
      background: #ff6347;
      color: #fff;
```

```
  padding: 2rem 0;
  text-align: center;
}

nav {
  display: flex;
  justify-content: center;
  background: #ff4500;
}

nav a {
  color: #fff;
  padding: 1rem 2rem;
  text-decoration: none;
  transition: background 0.3s ease;
}

nav a:hover {
  background: #ff6347;
}

section {
  padding: 2rem;
  max-width: 1200px;
  margin: auto;
}

.menu {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 1rem;
}

.menu-item {
  background: #f4f4f4;
  padding: 1rem;
  border-radius: 5px;
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.menu-item:hover {
  transform: translateY(-5px);
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
}
footer {
  background: #333;
```

```
        color: #fff;
        text-align: center;
        padding: 1rem 0;
        margin-top: 2rem;
      }
    </style>
  </head>
  <body>
    <header>
      <h1>Welcome to Delicious Bites</h1>
      <p>Your favorite place for mouth-watering meals!</p>
    </header>
    <nav>
      <a href="#home">Home</a>
      <a href="#menu">Menu</a>
      <a href="#about">About Us</a>
      <a href="#contact">Contact</a>
    </nav>
    <section id="menu">
      <h2>Our Menu</h2>
      <div class="menu">
        <div class="menu-item">
          <h3>Spaghetti Carbonara</h3>
          <p>A classic Italian pasta dish made with eggs, cheese, pancetta, and pepper.</p>
        </div>
        <div class="menu-item">
          <h3>Margherita Pizza</h3>
          <p>A simple yet delicious pizza topped with fresh tomatoes, mozzarella, and basil.</p>
        </div>
        <div class="menu-item">
          <h3>Caesar Salad</h3>
          <p>Crisp romaine lettuce tossed with Caesar dressing, croutons, and Parmesan cheese.</p>
        </div>
        <!-- Add more menu items as needed -->
      </div>
    </section>

    <!-- Add more sections like About Us and Contact as needed -->

    <footer>
      &copy; 2024 Delicious Bites. All rights reserved.
    </footer>
  </body>
  </html>
```

OUTPUT:



Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name

o Email

o Phone Number

o Department

o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

o Email format validation

o Phone number length validation

o Required field checks

Expected Outcome

• User-friendly and visually appealing form

• Real-time validation messages

• Prevention of invalid submissions’

PROMPT:
<!--Create an online registration form for SR University's technical fest with fields for Name, Email, Phone Number, Department, and Event Selection; style the form using CSS for readability and accessibility; implement JavaScript validation for email format, phone number length, and required field checks; ensure real-time validation messages and prevention of invalid submissions so the form is user-friendly, visually appealing, and fully functional.
-->

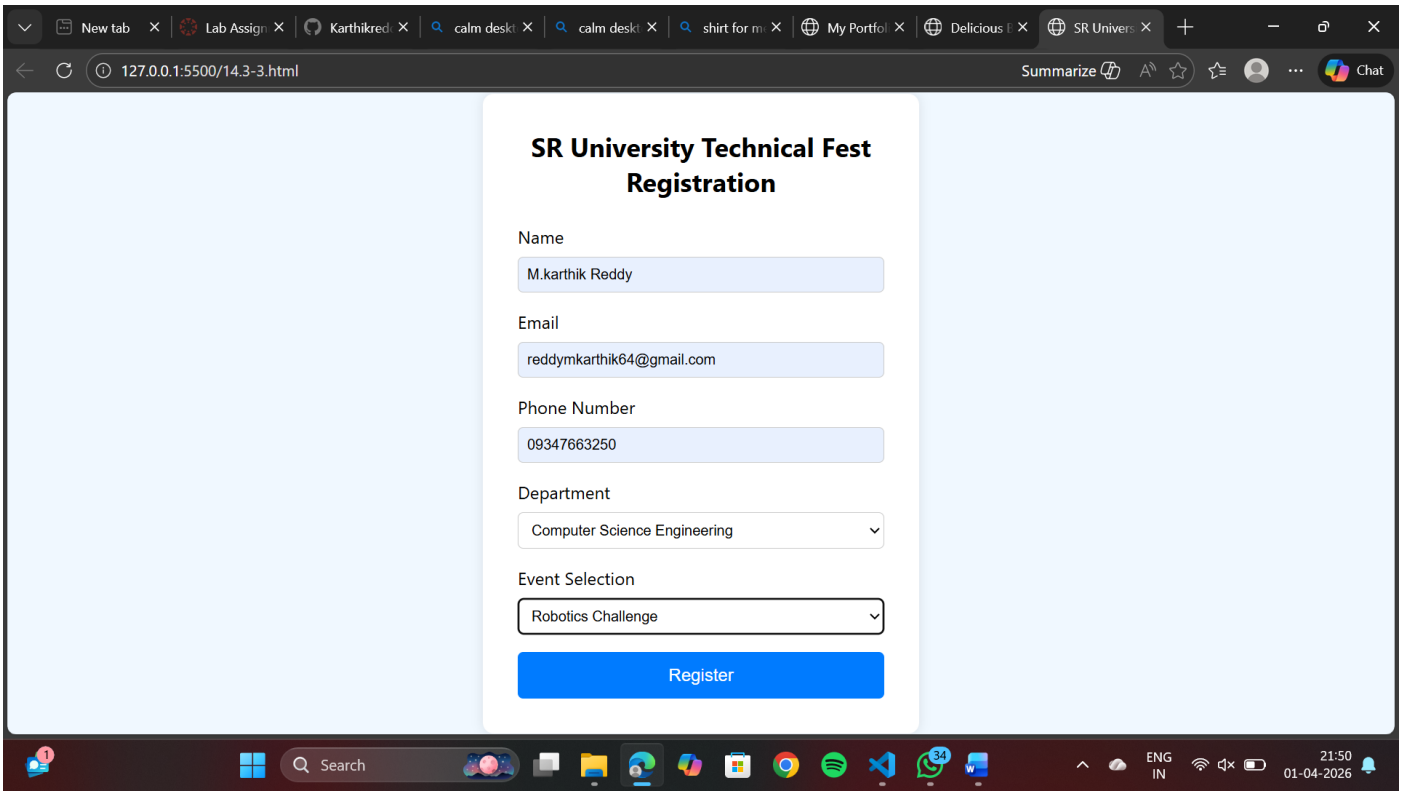
CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SR University Technical Fest Registration</title>
  <style>
    /* Basic Reset */
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f0f8ff;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }

    .registration-form {
      background: #fff;
      padding: 2rem;
      border-radius: 10px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      width: 100%;
      max-width: 400px;
    }

    .registration-form h2 {
      text-align: center;
      margin-bottom: 1.5rem;
```

```
        }

        .form-group {
            margin-bottom: 1rem;
        }

        .form-group label {
            display: block;
            margin-bottom: 0.5rem;
        }

        .form-group input,
        .form-group select {
            width: 100%;
            padding: 0.5rem;
            border: 1px solid #ccc;
            border-radius: 5px;
        }

        .error-message {
            color: red;
            font-size: 0.875rem;
            margin-top: 0.25rem;
        }

        .submit-btn {
            width: 100%;
            padding: 0.75rem;
            background-color: #007bff;
            color: #fff;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            font-size: 1rem;
        }

        .submit-btn:hover {
            background-color: #0056b3;
        }
    </style>
</head>
<body>
    <form class="registration-form" id="registrationForm">
        <h2>SR University Technical Fest Registration</h2>
        <div class="form-group">
```

```
        <label for="name">Name</label>
        <input type="text" id="name" name="name" required>
        <div class="error-message" id="nameError"></div>
      </div>
      <div class="form-group">
        <label for="email">Email</label>
        <input type="email" id="email" name="email" required>
        <div class="error-message" id="emailError"></div>
      </div>
      <div class="form-group">
        <label for="phone">Phone Number</label>
        <input type="text" id="phone" name="phone" required>
        <div class="error-message" id="phoneError"></div>
      </div>
      <div class="form-group">
        <label for="department">Department</label>
        <select id="department" name="department" required>
          <option value="">Select Department</option>
          <option value="CSE">Computer Science Engineering</option>
          <option value="ECE">Electronics and Communication Engineering</option>
          <option value="ME">Mechanical Engineering</option>
          <!-- Add more departments as needed -->
        </select>
        <div class="error-message" id="departmentError"></div>
      </div>
      <div class="form-group">
        <label for="event">Event Selection</label>
        <select id="event" name="event" required>
          <option value="">Select Event</option>
          <option value="coding">Coding Competition</option>
          <option value="robotics">Robotics Challenge</option>
          <option value="quiz">Technical Quiz</option>
          <!-- Add more events as needed -->
        </select>
        <div class="error-message" id="eventError"></div>
      </div>
      <button type="submit" class="submit-btn">Register</button>
    </form>

    <script src="./validation.js"></script> <!-- Link to your JavaScript validation file -->
  </body>
  </html>
```

OUTPUT:



Task 4: AI-Assisted E-Commerce Product Page
Scenario
A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.
Tasks
1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
o Cart item counter at the top-right corner
o Visual feedback when a product is added
Expected Outcome
• Functional product catalog
• Working cart counter
• Interactive user experience

PROMPT: <!--Create a basic e-commerce product page with a grid-based product catalog using HTML and CSS, implement an “Add to Cart” feature with JavaScript, include a cart item counter at the top-right corner, and provide visual feedback when a product is added; ensure the catalog is functional, the cart counter works correctly, and the overall user experience is interactive and responsive.
-->

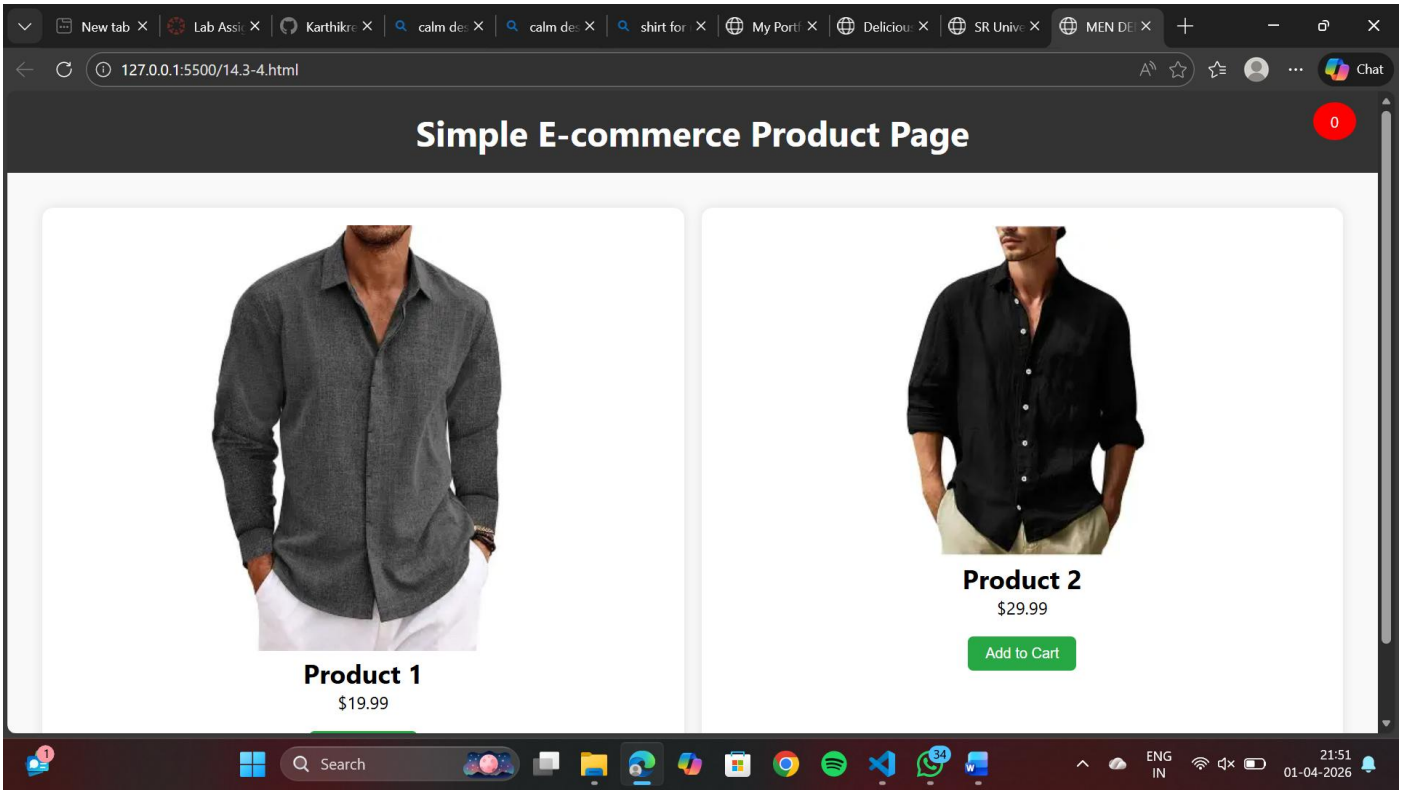
CODE:
<!DOCTYPE html>

```html
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MEN DEN</title>
  <style>
    /* Basic Reset */
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f9f9f9;
    }

    header {
      background: #333;
      color: #fff;
      padding: 1rem 0;
      text-align: center;
      position: relative;
    }

    .cart-counter {
      position: absolute;
      top: 10px;
      right: 20px;
      background: red;
      color: white;
      border-radius: 50%;
      padding: 0.5rem 1rem;
      font-size: 0.875rem;
    }

    .product-catalog {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
      gap: 1rem;
      padding: 2rem;
    }

    .product-card {
```

```
        background: #fff;
        padding: 1rem;
        border-radius: 10px;
        box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
        text-align: center;
      }

      .product-card img {
        max-width: 100%;
        height: auto;
      }

      .add-to-cart {
        margin-top: 1rem;
        padding: 0.5rem 1rem;
        background-color: #28a745;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
      }

      .add-to-cart:hover {
        background-color: #218838;
      }
    </style>
  </head>
  <body>
    <header>
      <h1>Simple E-commerce Product Page</h1>
      <div class="cart-counter" id="cartCounter">0</div>
    </header>

    <section class="product-catalog">
      <div class="product-card">
        <img src="https://th.bing.com/th/id/OIP.VrzKTGnYQnoLMbFQ-
xSQhgHaJj?w=202&h=260&c=7&r=0&o=7&dpr=1.5&pid=1.7&rm=3" alt="Product 1">
        <h2>Product 1</h2>
        <p>$19.99</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <div class="product-card">
        <img src="https://th.bing.com/th/id/OIP.CgPfFXY5w55LmfO0aMD-
oQHaHa?w=202&h=202&c=7&r=0&o=7&dpr=1.5&pid=1.7&rm=3" alt="Product 2">
        <h2>Product 2</h2>
```

```
        <p>$29.99</p>
        <button class="add-to-cart" onclick="addToCart()">Add to Cart</button>
      </div>
      <!-- Add more products as needed -->
    </section>

    <script>
      let cartCount = 0;

      function addToCart() {
        cartCount++;
        document.getElementById('cartCounter').textContent = cartCount;
        alert('Product added to cart!');
      }
    </script>

</html>
```

OUTPUT:



Task 5: Create a Responsive Web Page Layout

Instructions:

• Design a basic web page layout with a header, main content area, and footer using HTML and CSS.

• Use AI to assist in generating responsive CSS for different screen sizes.

• Ensure the layout is clean and visually organized.

Expected Output:

• A responsive web page with:

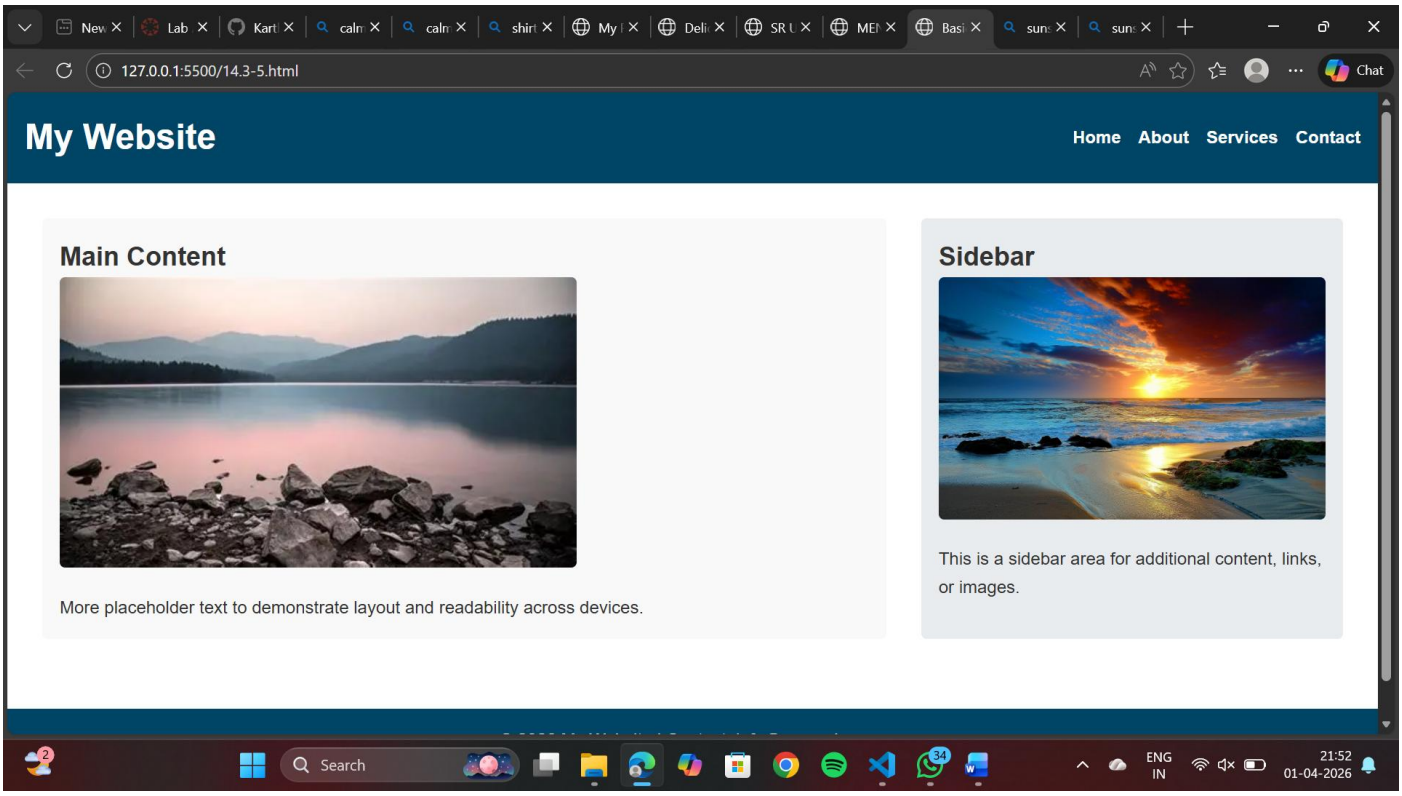
• Header with navigation links
• Main section with placeholder text/images
• Footer with copyright or contact info
• Layout adapts correctly to desktop, tablet, and mobile screen sizes.
Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

PROMPT:
<!--Design a basic web page layout with a header containing navigation links, a main content area with placeholder text and images, and a footer with copyright or contact info; use HTML and CSS with responsive styling for desktop, tablet, and mobile screen sizes; ensure the layout is clean, visually organized, and adapts correctly across devices with smooth readability and accessibility. English
-->

CODE:
```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Basic Responsive Layout</title>
 <style>
  /* Reset some defaults */
  * {
   margin: 0;
   padding: 0;
   box-sizing: border-box;
  }

  body {
   font-family: Arial, sans-serif;
   line-height: 1.6;
   color: #333;
  }

  header {
   background: #004466;
   color: #fff;
   padding: 1rem;
  }

  header nav {
   display: flex;
   justify-content: space-between;
```

```
  align-items: center;
}

header nav ul {
  list-style: none;
  display: flex;
  gap: 1rem;
}

header nav a {
  color: #fff;
  text-decoration: none;
  font-weight: bold;
}

main {
  padding: 2rem;
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 2rem;
}

main article {
  background: #f9f9f9;
  padding: 1rem;
  border-radius: 5px;
}

main aside {
  background: #e9ecef;
  padding: 1rem;
  border-radius: 5px;
}

main img {
  max-width: 100%;
  height: auto;
  border-radius: 5px;
  margin-bottom: 1rem;
}

footer {
  background: #004466;
  color: #fff;
  text-align: center;
```

```
    padding: 1rem;
    margin-top: 2rem;
  }

  /* Responsive styling */
  @media (max-width: 768px) {
    main {
      grid-template-columns: 1fr;
    }

    header nav {
      flex-direction: column;
      align-items: flex-start;
    }

    header nav ul {
      flex-direction: column;
      gap: 0.5rem;
    }
  }

  @media (max-width: 480px) {
    body {
      font-size: 0.9rem;
    }

    header, footer {
      text-align: center;
    }

    header nav {
      align-items: center;
    }
  }
 </style>
</head>
<body>
 <header>
  <nav>
    <h1>My Website</h1>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Services</a></li>
      <li><a href="#">Contact</a></li>
```

```
    </ul>
   </nav>
  </header>

  <main>
   <article>
    <h2>Main Content</h2>
    <img
src="https://tse4.mm.bing.net/th/id/OIP.Z4P3YrmiRaE8Spse9AGWxgHaEK?rs=1&pid=ImgDetMain&o
=7&rm=3" alt="Placeholder image">
    <p>More placeholder text to demonstrate layout and readability across devices.</p>
   </article>

   <aside>
    <h2>Sidebar</h2>
    <img src="https://via.placeholder.com/300x200" alt="Placeholder image">
    <p>This is a sidebar area for additional content, links, or images.</p>
   </aside>
  </main>

  <footer>
   <p>&copy; 2026 My Website | Contact: info@example.com</p>
  </footer>
</body>
</html>
```

OUTPUT: